J1A
DIN96 / male right-angle

A1		
A2		
A3	GND	
A4	+12V_BP	
A5	+12V_BP	
A6	GND	
A7	+5V_BP	
A8	+5V_BP	
A9	GND	
A10		
A11		
A12		
A13		
A14		
A15		
A16	HOST_SCL	
A17		ROW A
A18		
A19		
A20	HOST_RX	
A21		
A22		
A23		
A24	GND	
A25	+5V_BP	
A26	+5V_BP	
A27	GND	
A28	+12V_BP	
A29	+12V_BP	
A30	GND	
A31		
A32		

J1B
DIN96 / male right-angle

B1	GND	
B2	GND	
B3	GND	
B4	+12V_BP	
B5	+12V_BP	
B6	GND	
B7	+5V_BP	
B8	+5V_BP	
B9	GND	
B10	GND	
B11	GND	
B12	GND	
B13	GND	
B14	GND	
B15	GND	
B16	GND	
B17	GND	ROW B
B18	GND	
B19	GND	
B20	GND	
B21	GND	
B22	GND	
B23	GND	
B24	GND	
B25	+5V_BP	
B26	+5V_BP	
B27	GND	
B28	+12V_BP	
B29	+12V_BP	
B30	GND	
B31	GND	
B32	GND	

J1C
DIN96 / male right-angle

C1		
C2		
C3	GND	
C4	+12V_BP	
C5	+12V_BP	
C6	GND	
C7	+5V_BP	
C8	+5V_BP	
C9	GND	
C10		
C11		
C12		
C13		
C14		
C15		
C16	HOST_SDA	
C17		ROW C
C18		
C19		
C20	HOST_TX	
C21		
C22		
C23		
C24	GND	
C25	+5V_BP	
C26	+5V_BP	
C27	GND	
C28	+12V_BP	
C29	+12V_BP	
C30	GND	
C31		
C32		

Existing backplane interface: 12 V, 5 V, UART and I2C0.

No new pin assignments. All 24 V contacts remain unused.

J3
MOTOR 1

1	M1_BO1
2	M1_BO2
3	M1_AO2
4	M1_AO1

J4
MOTOR 2

1	M2_BO1
2	M2_BO2
3	M2_AO2
4	M2_AO1

J5
MOTOR 3

1	M3_BO1
2	M3_BO2
3	M3_AO2
4	M3_AO1

J6
MOTION ENCODERS

1	GND
2	+3V3
3	ENC_MOSI
4	ENC_MISO
5	ENC_SCK
6	ENC1_CS
7	ENC2_CS
8	ENC3_CS

J12
ROTATION ENCODER

1	+3V3
2	ROT_MISO
3	ROT_MOSI
4	ROT_SCK
5	ROT_CS
6	GND

J14
ROTATION MOTOR

1	ROT_OUT1
2	ROT_OUT2
3	ROT_OUT3
4	ROT_OUT4

J15
HEATER + TEMP

1	HEATER_5V
2	HEAT_LOW
3	+3V3
4	GND
5	TEMP_IN

J16
VACUUM / 5 V

1	+5V_LOAD
2	VAC_LOW

J17
THERMAL CUTOFF

1	+5V_LOAD
2	HEATER_5V

J2
SWD / RP2350

1	+3V3
2	RP_SWDIO
3	GND
4	RP_SWCLK
5	RP_RUN

J13
ENDSTOP

1	ROT_ENDSTOP
2	+3V3
3	GND

J19
TOOL PWM

1	GND
2	TOOL_PWM1
3	TOOL_PWM2
4	+3V3

U12
USBLC6-2SC6

USB_VBUS

C66
100nF 16V

USB_D+ 1 6 USB_D+
USB_D− 3 4 USB_D−
5 2

GND

USB_VBUS

R50
100kR 1%

USB_VBUS_SENSE

R51
150kR 1%

GND

J18
USB-C / SERVICE

VBUS	A4	USB_VBUS
CC1	A5	
CC2	B5	
D−	A7	USB_D−
D−	B7	
D+	A6	USB_D+
D+	B6	
SBU1	A8	
SBU2	B8	
SHIELD	SH	
GND	A1	

R49
5.1kR 1%

R48
5.1kR 1%

GND

Power
File: Power.kicad_sch

+5V_BP	+5V_BP
+12V_BP	+12V_BP
+3V3	+3V3
+5V_LOAD	+5V_LOAD
+12V_MOTOR	+12V_MOTOR
POWER_12V_SENSE	POWER_12V_SENSE

MotionController
File: MotionController.kicad_sch

+3V3	+3V3
RP_RUN	RP_RUN
RP_SWCLK	RP_SWCLK
RP_SWDIO	RP_SWDIO
USB_D+	USB_D+
USB_D−	USB_D−

MotionGPIO
File: MotionGPIO.kicad_sch

+3V3	+3V3
ENC1_CS	ENC1_CS
ENC2_CS	ENC2_CS
ENC3_CS	ENC3_CS
ENC_MISO	ENC_MISO
ENC_MOSI	ENC_MOSI
ENC_SCK	ENC_SCK
HEATER_PWM	HEATER_PWM
HOST_RX	HOST_RX
HOST_SCL	HOST_SCL
HOST_SDA	HOST_SDA
HOST_TX	HOST_TX
M1_A1	M1_A1
M1_A2	M1_A2
M1_B1	M1_B1
M1_B2	M1_B2
M2_A1	M2_A1
M2_A2	M2_A2
M2_B1	M2_B1
M2_B2	M2_B2
M3_A1	M3_A1
M3_A2	M3_A2
M3_B1	M3_B1
M3_B2	M3_B2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM
POWER_12V_SENSE	POWER_12V_SENSE
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_CS	ROT_CS
ROT_ENABLE	ROT_ENABLE
ROT_ENDSTOP	ROT_ENDSTOP
ROT_MISO	ROT_MISO
ROT_MOSI	ROT_MOSI
ROT_SCK	ROT_SCK
TEMP_IN	TEMP_IN
TOOL_PWM1	TOOL_PWM1
TOOL_PWM2	TOOL_PWM2
USB_VBUS_SENSE	USB_VBUS_SENSE
VAC_PWM	VAC_PWM

MotionDrivers
File: MotionDrivers.kicad_sch

+3V3	+3V3
+12V_MOTOR	+12V_MOTOR
M1_A1	M1_A1
M1_A2	M1_A2
M1_AO1	M1_AO1
M1_AO2	M1_AO2
M1_B1	M1_B1
M1_B2	M1_B2
M1_BO1	M1_BO1
M1_BO2	M1_BO2
M2_A1	M2_A1
M2_A2	M2_A2
M2_AO1	M2_AO1
M2_AO2	M2_AO2
M2_B1	M2_B1
M2_B2	M2_B2
M2_BO1	M2_BO1
M2_BO2	M2_BO2
M3_A1	M3_A1
M3_A2	M3_A2
M3_AO1	M3_AO1
M3_AO2	M3_AO2
M3_B1	M3_B1
M3_B2	M3_B2
M3_BO1	M3_BO1
M3_BO2	M3_BO2
MOTOR_EN	MOTOR_EN
MOTOR_PWM	MOTOR_PWM

ToolOutputs
File: ToolOutputs.kicad_sch

+3V3	+3V3
+5V_LOAD	+5V_LOAD
HEATER_PWM	HEATER_PWM
HEAT_LOW	HEAT_LOW
ROT_A1	ROT_A1
ROT_A2	ROT_A2
ROT_B1	ROT_B1
ROT_B2	ROT_B2
ROT_ENABLE	ROT_ENABLE
ROT_OUT1	ROT_OUT1
ROT_OUT2	ROT_OUT2
ROT_OUT3	ROT_OUT3
ROT_OUT4	ROT_OUT4
VAC_LOW	VAC_LOW
VAC_PWM	VAC_PWM

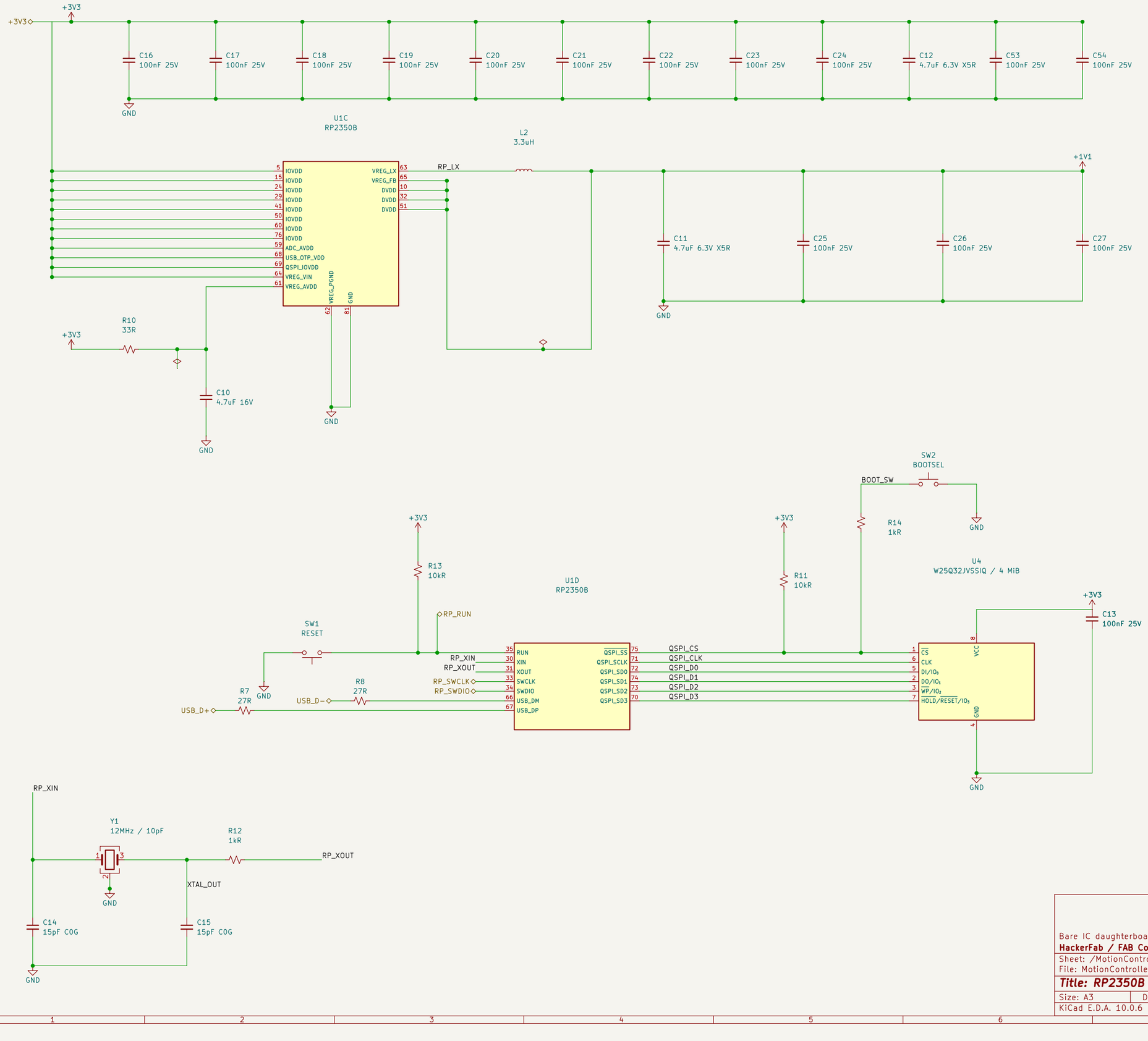
USB: data only; power the backplane before service.

Bare IC daughterboard / engineering review
HackerFab / FAB Computer

Sheet: /
File: FabNanoDaughterboard.kicad_sch

Title: FAB Nano

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/6



Bare IC daughterboard / engineering review

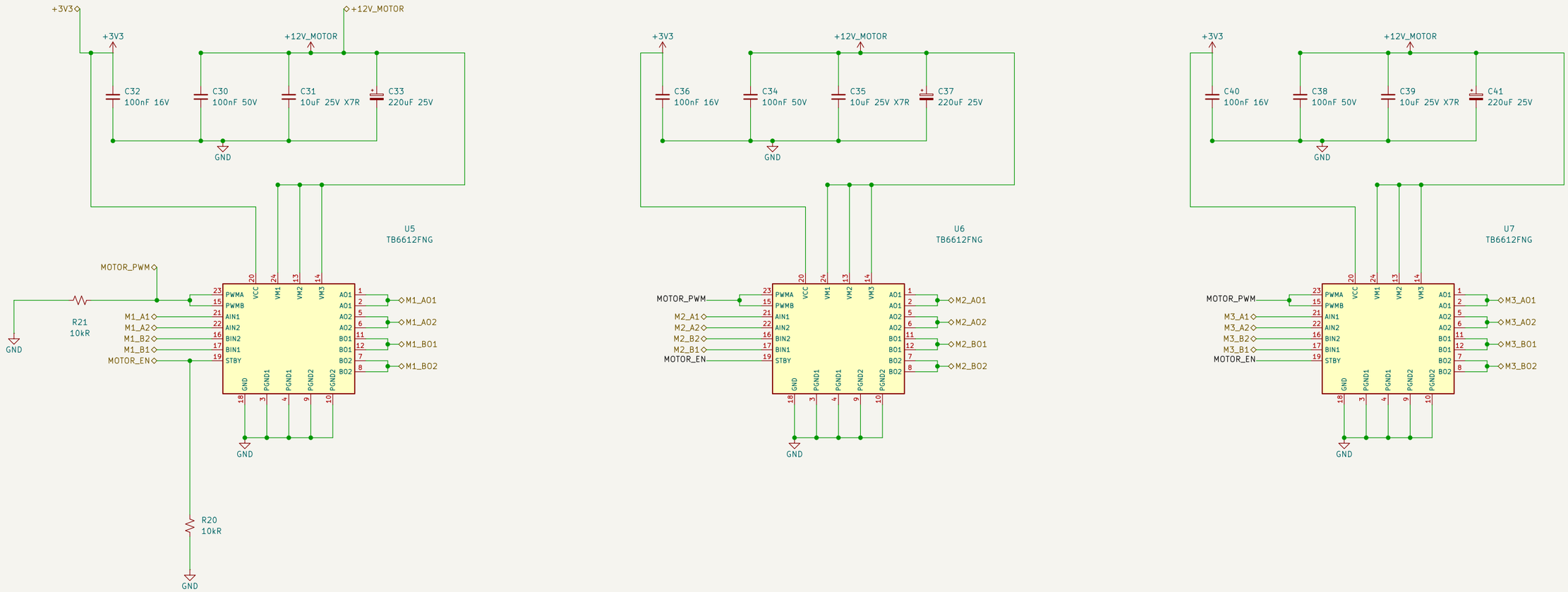
HackerFab / FAB Computer

Sheet: /MotionController/

File: MotionController.kicad_sch

Title: RP2350B

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/6



Bare IC daughterboard / engineering review

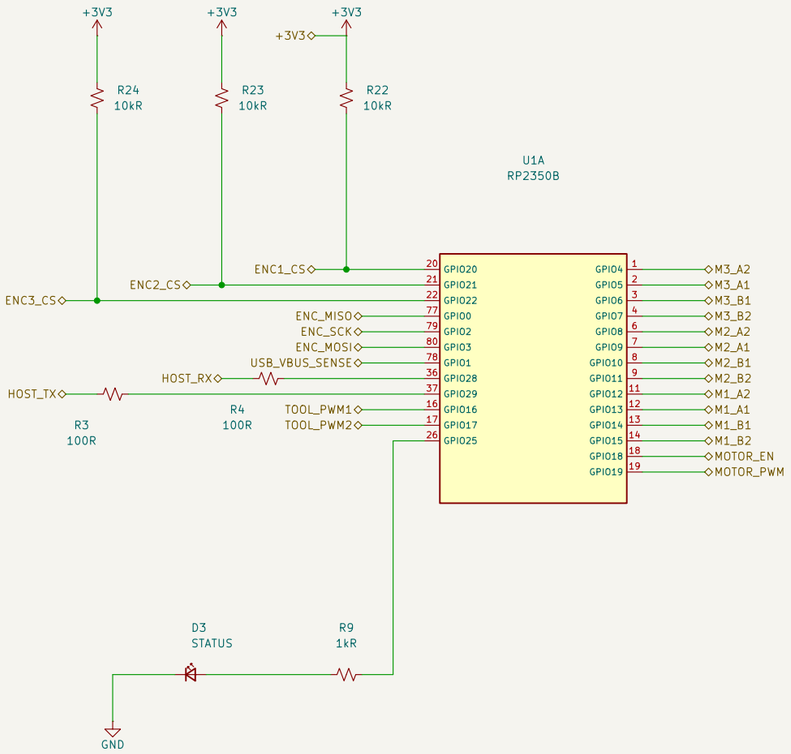
HackerFab / FAB Computer

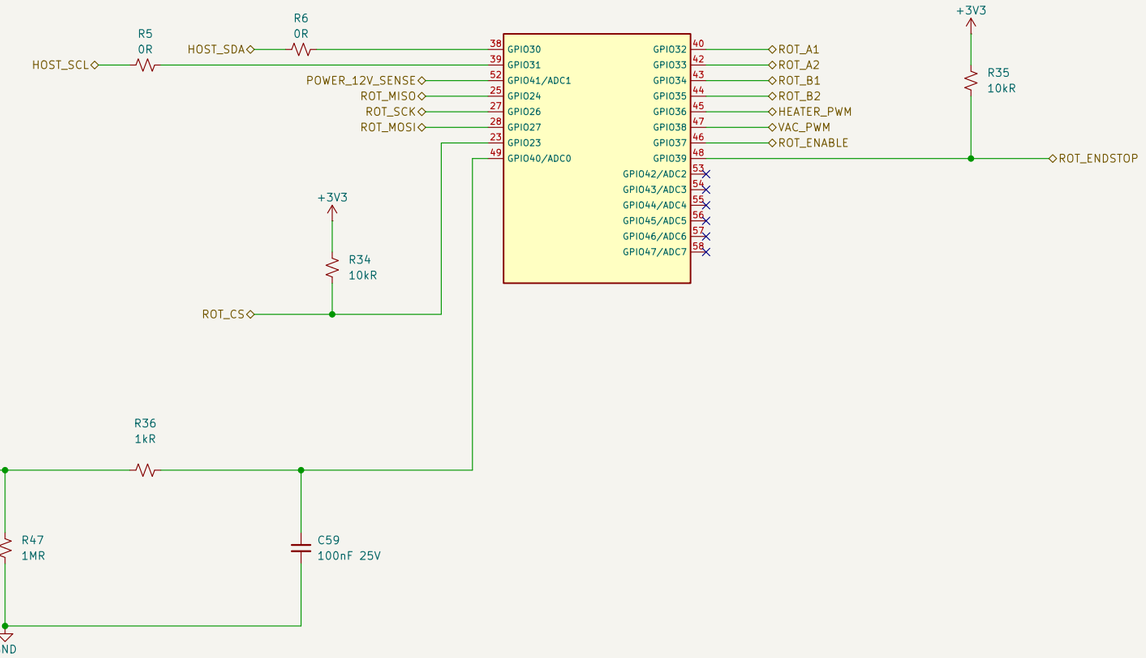
Sheet: /MotionDrivers/

File: MotionDrivers.kicad_sch

Title: Motion

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/6





One local controller: XYZ, rotation, temperature and vacuum.

Backplane UART and I2C0 contact assignments are unchanged.

Bare IC daughterboard / engineering review

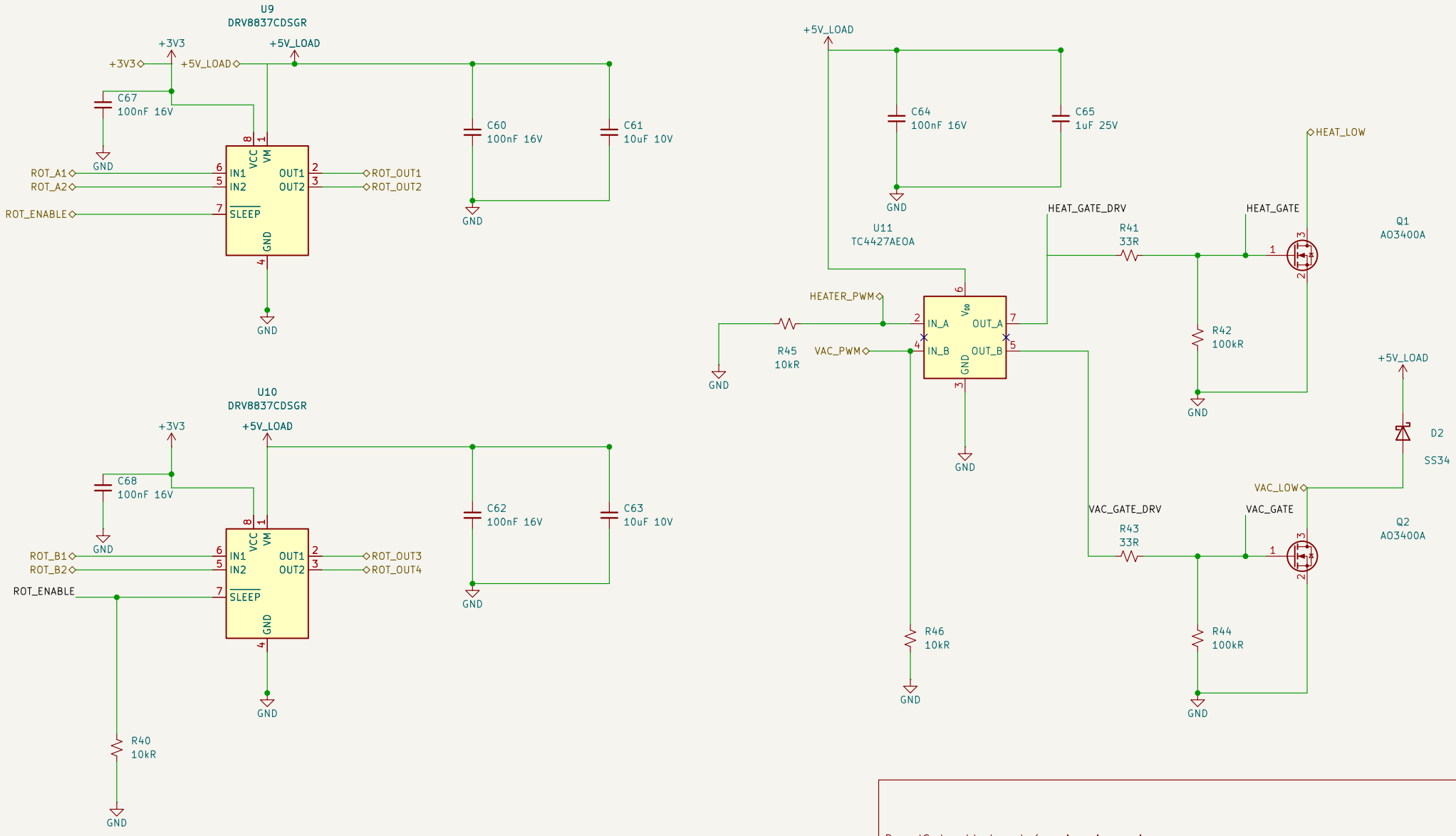
HackerFab / FAB Computer

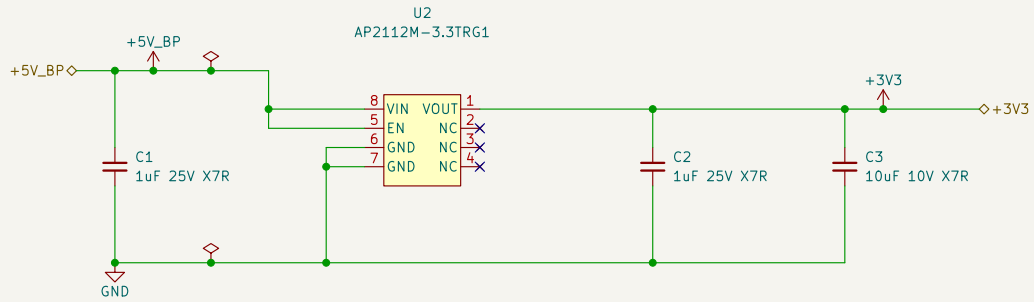
Sheet: /MotionGPIO/

File: MotionGPIO.kicad_sch

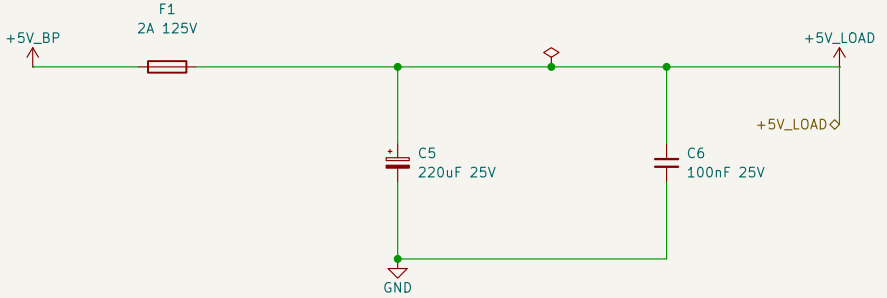
Title: RP2350B

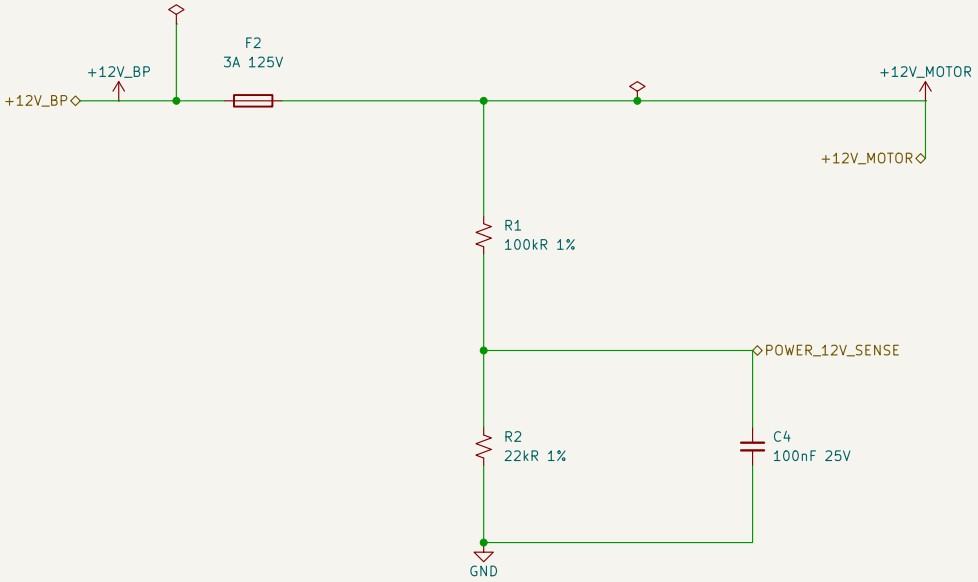
Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/6


U9
DRV8837CDSGR
+3V3
+5V_LOAD
C67
100nF 16V
C60
100nF 16V
C61
10uF 10V
GND
ROT_A1
ROT_A2
ROT_ENABLE
IN1
IN2
SLEEP
VCC
VM
OUT1
OUT2
ROT_OUT1
ROT_OUT2
U10
DRV8837CDSGR
C68
100nF 16V
C62
100nF 16V
C63
10uF 10V
ROT_B1
ROT_B2
ROT_OUT3
ROT_OUT4
R40
10kR
C64
100nF 16V
C65
1uF 25V
U11
TC4427AEOA
HEATER_PWM
VAC_PWM
IN_A
IN_B
OUT_A
OUT_B
R45
10kR
R46
10kR
HEAT_GATE_DRV
R41
33R
R42
100kR
HEAT_GATE
HEAT_LOW
Q1
AO3400A
D2
SS34
VAC_GATE_DRV
R43
33R
R44
100kR
VAC_GATE
VAC_LOW
Q2
AO3400A
Thermal cutoff J17 is in series with heater supply.
Headers and load wiring: top-level connector sheet.
Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /ToolOutputs/
File: ToolOutputs.kicad_sch
Title: Rotation, heater and vacuum
Size: A4
Date: 2026-11-06
Rev: A / DRAFT
KiCad E.D.A. 10.0.6
Id: 5/6



Logic only. No new backplane contact.





12 V absent: inhibit tool outputs.

Bare IC daughterboard / engineering review

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Sheet: /Power/

File: Power.kicad_sch

Title: Backplane rails and interfaces

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 6/6